- $u_1 + u_3 = 0_{\mathbb{R}^3} \Rightarrow$ -
- $-u_2 + 1u_4 = 0_{\mathbb{R}^3}$
- $\alpha u_1 + \beta u_2 + \gamma u_3 + \lambda u_4 = 0_{\mathbb{R}^3} = (0,0,0)$

$\Rightarrow \alpha(1,2,1) + \beta(0,2,1) + \gamma(-1,-2,-1) + \lambda(0,1,\frac{1}{2}) = 0_{\mathbb{R}^3}$

- $u_1 + u_3 = 0_{\mathbb{R}^3} \Rightarrow u_1 = -u_3 \Rightarrow (u_1, u_2, u_3, u_4)$ liée.
- $F = \text{vect}(u_1, u_2, u_3)$
- (u_1, u_2, u_3) est génératrice de F

Et $u_1 + u_3 = 0_{\mathbb{R}^3} \Rightarrow (u_1, u_2)$ génératrice de F

$(u_1, u_2, -\cancel{u_1})$

- $F = \text{vect}(u_1, u_2, \cancel{u_3}) = \text{vect}(u_1, u_2)$
car $u_3 = -u_1$

$\alpha u_1 + \beta u_2 = 0_{\mathbb{R}^3} \Rightarrow \alpha(1,2,1) + \beta(0,2,1) = 0_{\mathbb{R}^3}$

$\Rightarrow \begin{cases} \alpha = 0 \\ 2\alpha + 2\beta = 0 \\ \alpha + \beta = 0 \end{cases} \Rightarrow \alpha = \beta = 0 \Rightarrow (u_1, u_2)$ est libre

$\Rightarrow (u_1, u_2)$ est une base de F.

- $F = \text{vect}(u_1, u_2) = \{x u_1 + y u_2 / x, y \in \mathbb{R}\}$
$= \{x(1,2,1) + y(0,2,1) / x, y \in \mathbb{R}\}$
$= \{(x, 2x + 2y, x + y) / x, y \in \mathbb{R}\}$

$$F = \{(x, \underline{2x+2y}; x+y) / x, y \in \mathbb{R}\}$$

$$= \{(x, 2(x+y); x+y) / x, y \in \mathbb{R}\}$$

$$= \{(x', y', z') / x' \in \mathbb{R} \text{ et } y' = 2z'\}$$
y', z'

$$= \{(x, y, z) / x \in \mathbb{R} / y = 2z\}$$

- $u_1 = - u_3 = \alpha u_3 / \alpha = -1$
- $F = \{(x, y, z) / x \in \mathbb{R} \text{ et } y = 2z\}$
- $G = \{(x, y, z) / x = 0 \text{ et } y = 2z\}$
- $G \subset F$ car $(0, 2z, z) \in G$; $(0, 2z, z) \in F$

$$G = \{(x,y,z) \in \mathbb{R}^3 / x = 0 \text{ et } y - 2z = 0\}$$
$$= \{(0, 2z, z) \in \mathbb{R}^3 / z \in \mathbb{R}\}$$
$$= \{z(0,2,1) / z \in \mathbb{R}\} = \mathrm{vect}(u_1)$$

G est s.e.v de $\mathbb{R}^3$.

- $A \cap B$ A

$A \cup B$ B

$$A \subset B \Rightarrow \begin{cases} A \cap B = A \\ A \cup B = B \end{cases}$$

si $A \subset B$

$A \cap B = A \subset B$

$$\boxed{E \subset F \Rightarrow \dim E \leq \dim F}$$

• On a $G \subset F$ donc $F \cap G = G$

Alors $\dim(F \cap G) = \dim G = 1$

• $G \subset F \Rightarrow G \subset G + F \subset F + F = F$

$\Rightarrow F \subset G + F = G$ absurde

• $\dim(F+G) = \dim F + \dim G - \dim \overset{G}{\overset{\shortparallel}{(F \cap G)}}$

$= 2 + 1 - 1 = 2$

$\Rightarrow F + G \neq \mathbb{R}^3 \longrightarrow \dim \mathbb{R}^3 = 3$

(Autre Méthode $G + F = F$ car $G \subset F$

$\Rightarrow G + F \neq \mathbb{R}^3$

$\dim F = 2$)

$\cdot\ u_4 \notin H$ car $1 - 2\left(\frac{1}{2}\right)^2 = \frac{1}{2} \neq 0$

$\Longrightarrow$ u_4 et $2u_4 \notin H$.

$2u_4 = (0, 2, 1) \in H$ et $u_4 \notin H$

Alors H n'est pas un s.e.v.

F s.e.v de $E \Longleftrightarrow$ $\begin{cases} \cdot\ \forall x, y \in F,\ x + y \in F \\ \cdot\ \forall x \in F,\ \forall \alpha \in \mathbb{R} \\ \quad \alpha x \in F \end{cases}$

$\neq \emptyset$

$0_E \in F$.

$f(0_E) = 0_F$

$\Longleftrightarrow \begin{cases} \forall x, y \in F,\ \forall \alpha \in \mathbb{K} \\ \alpha x + y \in F \end{cases}$

$\Longleftrightarrow F = \text{vect}(u_1, u_2, \ldots, u_n)$

$u = (0,2,1) \in H$ car $2 - 2(1)^2 = 0$

$v = (0,2,-1) \in H$ // $2 - 2(-1)^2 = 0$

$u + v = (0,4,0) \notin H$ car $4 - 2(0) \neq 0$

- $f : \mathbb{R}_1[X] \longrightarrow \mathbb{R}_2$

$P = a_0 + a_1 X \longmapsto f(P) = (a_0 - a_1 ; a_0 + 2a_1)$

- $f(P + Q) = f(a_0 + a_1 X + a_0' + a_1' X)$

$= f(\overbrace{a_0 + a_0'}^{a_0''} + \overbrace{(a_1 + a_1')}^{a_1''} X)$

$= (a_0 + a_0' - a_1 - a_1' ; a_0 + a_0' + 2(a_1 + a_1'))$

$= (a_0 - a_1 ; a_0 + 2a_1) + (a'_0 - a'_1 ; a'_0 + 2a'_1)$

$= f(P) + f(Q)$

$P = a_0 + a_1 X$

$\cdot\ f(\alpha P) = f(\underbrace{a_0 \alpha} + \alpha a_1 X)$

$= (\alpha a_0 - \alpha a_1 ; \alpha a_0 + 2\alpha a_1)$

$= \alpha (a_0 - a_1 ; a_0 + 2a_1)$

$= \alpha f(P)$

Donc f est une a.l.

$$g : \mathbb{R}^2 \longrightarrow \mathbb{R}$$
$$(x, y) \longmapsto x^2 + y^2$$

$$\underline{X = (1, 0)} \qquad \alpha = 2$$

$$g(2X) = g(2, 0) = 2^2 + 0^2 = 4$$

$$2\, g(X) = 2\, g(1, 0) = 2(1^2 + 0^2) = 2$$

$$\Longrightarrow \quad g(2X) \neq 2\, g(X)$$

$$f : \mathbb{R}_1[X] \longrightarrow \mathbb{R}^2$$

$$P = a_0 + a_1 X \longrightarrow f(P) = (a_0 - a_1, a_0 + 2a_1)$$

$$\operatorname{Ker} f = \{ P = a_0 + a_1 X \,/\, (a_0 - a_1, a_0 + 2a_1) = 0_{\mathbb{R}^2} \}$$

$$= \left\{ P = a_0 + a_1 X \,/\, \begin{cases} a_0 - a_1 = 0 \\ a_0 + 2a_1 = 0 \end{cases} \right\}$$

$$= \left\{ P = a_0 + a_1 X \,/\, \begin{cases} a_0 = a_1 \\ a_1 = 0 \end{cases} \right\}$$

$$= \{ 0 \}$$

$$\dim \mathbb{R}_1[X] = 2 = \dim \operatorname{Im} f + \dim \operatorname{Ker} f$$

$f : E \longrightarrow F$

- $\forall y \in F, \exists x \in E \,/\, f(x) = y$

$\Leftrightarrow f$ surjective

$\dim \operatorname{Im} f = 2 = \dim \mathbb{R}^2$

$\operatorname{Im} f = \mathbb{R}^2 \Rightarrow$ surjective

$f : \mathbb{R}[X] \longrightarrow \mathbb{R}^2$

$\forall (a;b) \in \mathbb{R}^2 ; \exists P \in \mathbb{R}[X] \,/\, f(P) = (a,b)$